

LOVEPREET SIDHU

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SUMMARY OF QUALIFICATIONS

- Bachelor of Technology in Information Technology (Network Administration & Security) at Kwantlen Polytechnic University specializing in **system troubleshooting, enterprise infrastructure, and distributed systems** — with hands-on experience diagnosing authentication failures, network anomalies, and security incidents across **Linux** and **Windows Server** environments.
- Proficient in **databases** (MySQL, MongoDB, SQL), **operating systems** (Linux Ubuntu/CentOS, Windows Server), **networking protocols** (TCP/IP, DNS, HTTP), and **security/authentication** technologies (SAML, 802.1X, RADIUS, SSL)
- **AWS Certified Solutions Architect** and **Cloud Practitioner** with project experience in **cloud architecture**, containerization (Docker), and enterprise identity infrastructure; experienced in Java, Python, and C++ with strong **object-oriented programming** and **debugging skills**.
- Strong communicator and collaborative team player, developed through coordinating urgent requests with medical staff in a fast-paced clinical environment at Vancouver Coastal Health and 3+ years of prior customer-facing technical roles (Bell, Telus) — bringing proven skills in **communication, customer service**, and **operational efficiency** that complement strong technical documentation abilities demonstrated across security, networking, and cloud infrastructure projects.

TECHNICAL SKILLS

Databases & ERP: MySQL, MongoDB, SQL, Odoo (ERP)

Operating Systems: Linux (Ubuntu/CentOS), Windows Server (AD DS, GPO, NPS), Command Line (Bash)

Networking & Protocols: TCP/IP, DNS, HTTP, Cisco IOS, VLANs, VPNs, OSPF, BGP

Security & Authentication: SAML, 802.1X, RADIUS, SSL, Firewall Policy, Nmap, Wireshark

Languages & OOP: Java, Python (Boto3), C++, JavaScript, PHP, Bash

Cloud & Virtualization: AWS (VPC, EC2, S3, IAM, CloudFormation), Docker, VMware, Hyper-V

Monitoring & Tools: Splunk, Wazuh, SIEM, OSSEC, Git, Postman, RESTful APIs

EDUCATION

Kwantlen Polytechnic University – Bachelor of Technology – Information Technology

Sep 2022 – Dec 2026 (Expected)

Specialization: Network Administration and Security

Coursework: INFO 3240 (ERP Systems), INFO 3171 System Security, INFO 4110 Cloud Computing

Certifications: AWS Certified Solutions Architect – Associate (November 2024) · AWS Certified Cloud Practitioner (February 2024)

PROJECTS

STaaS Cloud-Based Smart Drive

December 2025

Cloud Computing

- Designed and troubleshot secure AWS infrastructure including VPC segmentation, NAT gateways, and bastion hosts; diagnosed and resolved connectivity issues across distributed cloud environments.
- Developed and debugged a Spring Boot backend integrated with AWS S3; automated cloud operations and file lifecycle management using Python (Boto3).
- Containerized services with Docker to ensure scalable, consistent deployment; investigated and resolved container networking and dependency issues across distributed environments.
- Stack: AWS (VPC, EC2, S3, IAM), Spring Boot, Docker, MongoDB, Python.

File Integrity Monitoring Tool & Security/Forensics Lab

March 2025

System Security & Digital Forensics

- Built a Java-based FIM tool using SHA-256 hashing to detect and investigate unauthorized system changes in real-time; configured OSSEC HIDS on Linux VMs to centralize log monitoring, rootkit detection, and automated alerting.
- Investigated network anomalies and security incidents using Wireshark packet analysis, performing root cause analysis and documenting findings to ensure operational visibility and data integrity.
- Tools: Java, Linux (Ubuntu), OSSEC, Wireshark, TCP/IP, DNS, Autopsy.

Enterprise Authentication Infrastructure

February 2026

Security of Wireless Systems

- Deployed and troubleshot an enterprise identity backend using Windows Server 2022, Active Directory (AD DS), and RADIUS; diagnosed authentication failures and resolved connectivity issues across multi-system environments.
- Investigated and analyzed authentication logs following SRE-style incident response workflows to perform root cause analysis on system failures and access anomalies.

- Researched SAML SSO and identity federation protocols to understand how application-level authentication complements network-level access control implemented via RADIUS — broadening understanding of end-to-end enterprise identity architecture.
- Tools: Active Directory, RADIUS, VMware Fusion, Windows Server 2022, 802.1X.

EXPERIENCE

Vancouver Coastal Health – Porter

October 2022 – Present

Vancouver, BC

- Transport patients of varying acuity and complexity between departments using stretchers, wheelchairs, and beds; coordinate with nursing and technical staff to prioritize urgent requests and ensure timely, safe patient movement.
- Manage and maintain medical supplies, equipment, and storage areas across a high-volume acute care environment; deliver specimens, blood products, and critical materials with accuracy and time sensitivity.
- Communicate effectively with multidisciplinary clinical teams to assess priorities, adapt to shifting operational demands, and minimize workflow disruptions in a fast-paced hospital setting.

United Parcel Service (UPS) – Delivery Driver

October 2021 – February 2022

Delta, BC

- Executed time-sensitive deliveries using logistics software and real-time tracking systems to meet strict deadlines
- Optimized delivery routes independently to maximize efficiency and maintain vehicle safety standards

BELL (Authorized Dealer)– Sales Representative

April 2018 – August 2021

Surrey, BC

- Diagnosed and resolved customer-reported technical issues including mobile device setup failures, software errors, and network connectivity problems across a high volume of daily client interactions.
- Translated complex technical concepts — including 5G network architecture and hardware specifications — into clear, accessible explanations for non-technical customers to drive informed decision-making.
- Maintained high customer satisfaction by systematically troubleshooting connectivity and hardware issues, identifying root causes, and delivering timely resolutions under a fast-paced retail environment.

TELUS (Authorized Dealer)– Field Sales Representative

April 2017 – March 2018

Surrey, BC

- Advised customers on fiber network upgrades and provided setup support for home connectivity solutions
- Educated clients on product features and installation processes, simplifying technical concepts for non-technical users

VOLUNTEER

Cystic Fibrosis Foundation – Event Support Volunteer

May 2025

Burnaby, BC

- Supported logistics coordination and participant assistance to raise awareness and funds for Cystic Fibrosis research